

Ananth Shreekumar

Ph.D. Student, Department of Computer Science
Purdue University, West Lafayette, IN

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EDUCATION

Doctor of Philosophy in Computer Science

Purdue University

- Advisors: Dr. Z. Berkay Celik, Dr. Dongyan Xu

Present

West Lafayette, IN

Master of Science in Computer Science

Purdue University

- GPA : 3.93 / 4.0

Dec 2023

West Lafayette, IN

Integrated Master of Technology in Computer Science and Engineering

International Institute of Information Technology Bangalore (IIIT-B)

- GPA : 3.92 / 4.0

Jul 2021

Bangalore, India

EXPERIENCE

American Express

Software Engineer, Enterprise Architecture Team

- Integrated machine learning to internal logging and monitoring framework to detect anomalies from logs.
- Implemented a pipeline to perform automated log analysis on real-time log data.

Aug 2021 - Dec 2021

Bangalore, India

Siemens Healthineers

Technical Intern, Center for Innovation in Diagnostics Team

- Designed a deep learning pipeline to predict Sepsis onset in ICU patients from bedside measurements.
- Model performance exceeded 87% on accuracy, F1-score, specificity, and sensitivity metrics.

Jan 2021 - Jul 2021

Bangalore, India

PUBLICATIONS

Conference Publications

- C3 Ananth Shreekumar, JyunJhu Syu, Muslum Ozgur Ozmen, Dongyan Xu, Z. Berkay Celik. A Self-Explainable Deep Architecture for Security Applications. *Conference on Game Theory and AI for Security*. 2026.
- C2 Devin Ersoy*, Brandon Lee*, Ananth Shreekumar, Arjun Arunasalam, Muhammad Ibrahim, Antonio Bianchi, and Z. Berkay Celik. Investigating the Impact of Dark Patterns on LLM-Based Web Agents. *IEEE Symposium on Security and Privacy*. 2026.
- C1 Ananth Shreekumar*, Biswesh Mohapatra*, and Shrisha Rao. Incorporating Autonomous Bargaining Capabilities into E-Commerce Systems. *ACM International Conference on Intelligent Virtual Agents*. 2020.

Journal Articles

- J1 Ankit Gupta, Ruchi Chauhan, Saravanan G, and Ananth Shreekumar. Improving Sepsis Prediction in Intensive Care with SepsisAI: A clinical decision support system with a focus on minimizing false alarms. *PLOS Digital Health*. 2024.

Workshop Publications

- W2 Edoardo Allegrini, Ananth Shreekumar, and Z. Berkay Celik. Formalizing the Safety, Security, and Functional Properties of Agentic AI Systems. *Agents in the Wild: Safety, Security, and Beyond Workshop, International Conference on Learning Representations*. 2026.
- W1 Tarun Dutt, GNS Prasanna, TR Dastidar, and Ananth Shreekumar. Towards Artifact Rejection in Microscopic Urinalysis. *Medical Imaging Meets NeurIPS Workshop, Neural Information Processing Systems*. 2019.

Preprints

- P3 Doguhan Yeke*, Elif Su Temirel*, Ananth Shreekumar*, Brandon Lee, Dongyan Xu, Z. Berkay Celik. The Yes-Man Syndrome: Benchmarking Abstention in Embodied Robotic Agents. *arXiv preprint arXiv:2605.20544*. 2026.
- P2 Zengxiang Lei, Ananth Shreekumar, Jonathan Rosenthal, Ruoyu Song, Alvaro A. Cardenas, Daniel J. Fremont, Dongyan Xu, Satish Ukkusuri, Z. Berkay Celik. Stable GFlowNets with TV Monitoring and Probabilistic Guarantees. *arXiv preprint arXiv:2605.01729*. 2026.
- P1 Mohammad Atif Quamar, Mohammad Areeb, Nishant Sharma, Ananth Shreekumar, Jonathan Rosenthal, Muslum Ozgur Ozmen, Mikhail Kuznetsov, Z. Berkay Celik. Adaptive Blockwise Search: Inference-Time Alignment for Large Language Models. *arXiv preprint arXiv:2510.23334*. 2025.

SERVICE

External Reviewer

SIGKDD (2026, 2027), NDSS (2025, 2026), IEEE S&P (2025, 2026), CCS (2025), NeurIPS (2025), AAAI (2026), ICLR (2026)

Outreach

- Presented my research to about 20 high school students as part of a 4-H event. 2024, 2025.

Graduate Teaching Assistant

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| • CS361 Great Issues in Computer Science, Purdue University. | Spring 2025 |
| • CS529 Security Analytics, Purdue University. | Fall 2024 |
| • CS240 Programming in C, Purdue University. | Fall 2023 |
| • CS352 Compilers: Principles and Practice, Purdue University. | Spring 2023 |
| • MA165 Calculus I, Purdue University. | Fall 2022 |

Guest Lecturer

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| • CS197 Freshman Honors Seminar, Purdue University. | Spring 2026 |
| • CS529 Security Analytics, Purdue University. | Fall 2025 |
| • CS361 Great Issues in Computer Science, Purdue University. | Spring 2025 |
| • CS529 Security Analytics, Purdue University. | Fall 2024 |

REFERENCES

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Dongyan Xu

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